

Efficient Speed and Voltage Regulation of BLDC Motor Drive for EV Applications using a Multi-Device Interleaved DC–DC Bidirectional Converter with TIFDNFD–SFOA Controller

Abdullah Alwabli^{1,}*

¹ Department of Electrical Engineering, College of Engineering and Computing in Al-Qunfudhah,

Umm al-Qura University, Mecca, Saudi Arabia

*Corresponding Email: aswabli@uqu.edu.sa

Abstract - Electric vehicles (EVs) are emerging as a key solution for sustainable transportation, where efficient motor drives and power electronic interfaces play a crucial role in ensuring reliable performance. In modern EV systems, simultaneous regulation of motor speed and DC-link voltage is essential for stable inverter operation and improved drive efficiency; however, many existing studies primarily focus on speed control while giving limited attention to coordinated voltage regulation. To address this challenge, this research proposes the design, simulation, and analysis of a Multi-Device Interleaved DC–DC Bidirectional Converter (MDIBC) integrated with a Tilted Integral Fractional Derivative with Filter plus Fractional Derivative (TIFDNFD) controller for enhanced voltage and speed regulation of Brushless DC (BLDC) motor drives. The proposed MDIBC employs phase-shifted PWM excitation to minimize ripple current, enhance current sharing among interleaved inductors, and maintain a stable DC link voltage under varying load conditions. To achieve superior dynamic response and robust voltage and speed regulation, the TIFDNFD controller parameters are optimized using the Superb Fairy-wren Optimization Algorithm (SFOA), which improves convergence speed and tuning accuracy. Simulation results obtained in the MATLAB/Simulink environment demonstrate superior performance

over conventional controllers, achieving minimal peak time of 0.0484s, settling time of 0.12 s and minimum overshoot of 3.3%. The proposed system proves highly suitable for advanced EV and industrial drive applications.

Keywords: TIFDNFD Controller, Superb Fairy-wren Optimization Algorithm, Electric vehicle, Brushless DC Motor, Multi-Device Interleaved DC–DC Bidirectional Converter, Voltage and Speed Regulation.

Nomenclature

EVs	Electric Vehicles
MDIBC	Multi-Device Interleaved Dc–Dc Bidirectional Converter
TIFDNFD	Tilted Integral Fractional Derivative With Filter Plus Fractional Derivative
BLDC	Brushless DC
SFOA	Superb Fairy-Wren Optimization Algorithm
RBFNN	Radial Basis Function Neural Networks
ANN	Artificial Neural Networks
PID	Proportional-Integral-Derivative
FOSMC	Fractional Order Sliding Mode Controller
ITAE	Lower Integral Time Absolute Error
SEPIC	Single Ended Primary Inductance Converters

PEMFCs	Proton Exchange Membrane Fuel Cells
MPPT	Maximum Power Point Tracking
UFENN	Unified Firefly Ersatz Neural Network
VSI	Voltage Source Inverter
PI	Proportional Integral
BL-CC	Bridgeless Canonical Cell
PFC	Power Factor Correction
IGBT	Insulated Gate Bipolar Transistor
CCM	Continuous Conduction Mode
DIC	Discontinuous Inductor Current
PAM	Pulse Amplitude Modulation
LDC	Low-Voltage Dc-Dc Converter
ZVS	Zero-Voltage Switching
SCC	Switch-Controlled Capacitor
PV	Solar Photovoltaic
FOC	Field-Oriented Control
HCC	Hysteresis Current Control
PWM	Pulse Width Modulation

1. Introduction

The need for dependable and highly efficient power conversion and motor control technologies has increased because EVs and renewable energy sources are developing quickly [1]. Internal combustion engines are now the primary source of mobility for vehicles. Since fuel supplies are running out more quickly than anticipated, EVs are becoming essential [2]. EV deployment in the consumer market is crucial, and automakers are

investigating efficient technologies for EVs [3]. In recent years, the drive train technology used in the automotive industry underwent a revolution [4]. The development of EV technology, which uses an electric motor and battery, demonstrated that these vehicles are both more environmentally friendly and more efficient than internal combustion engines [5]. An electric motor powers an EV's wheels. The battery is one of the primary power sources for an EV's motor while it is in motion [6]. The battery needs to be recharged after usage. As a result, the DC-DC converter needs to be more effective [7]. Numerous DC-DC converter topologies have been researched in recent years [8]. Among these, the MDIBC offers a dependable architecture, low electromagnetic interference, bidirectional operation, and input and output voltage ripples [9]. Vehicles with high-power motors can use this converter [10]. Numerous DC-DC converter topologies are utilized in hybrid EVs and the topic of fast charging station research is being thoroughly investigated [11]. It was decided that the most suitable case of the EV application was the interleaved boost converter for several devices, in which MDIBC was effective in producing low ripples of voltage and current [12].

In the aviation sector, BLDC motors are now essential, especially for the advancement of EVs [13]. The motors are most suited to numerous applications in EV systems due to number of advantages over conventional motors [14]. Being the primary propulsion system in electric propulsion is one of the major uses for BLDC motors. BLDC motors have a high torque and efficiency, which enables EVs to move faster and farther [15]. Also, BLDC motors offer controlled and reliable motion as actuators in vital systems like cargo doors, landing gear, and flight control surfaces [16]. In order to ensure passenger comfort, Additionally, they are utilized for ecological control systems to manage cabin temperature and ventilation [17]. BLDC motors are also essential for power generation, fuel pumps, and cooling systems, all of which improve the vehicle's overall performance and energy management [18]. Using BLDC motors in automotive systems improves performance and

efficiency while lowering dependency on conventional mechanical and hydraulic systems. Electronic commutators are necessary for these motors, and they need details regarding the rotor position in order to use a voltage source inverter to accomplish commutation [19]. BLDC motors have very important speed-torque characteristics and don't require brush placement, which lowers copper and eddy current losses. There are no excitation losses due to its permanent magnet excitation which raise the BLDC motor's efficiency to a sufficiently high level [20].

Accurate control strategies that can manage both speed and DC-link voltage are necessary for BLDC motor drives. Recent studies have developed intelligent control methods such as RBFNN, ANN and hybrid machine-learning-based controllers. However, these methods are computationally complex and require a lot of training, and they are not robust under abrupt and nonlinear load variations. Fuzzy logic, sliding mode control, and proportional-integral-derivative (PID) are examples of conventional controllers that function well in steady state but suffer from overshoot, slow convergence, parameter sensitivity, and decreased performance in transient disturbances. Complex fractional parameter tuning, which raises computational load and could result in chattering close to the sliding surface, is the main problem of FOSMC. The majority of optimization algorithms have issues with multidimensional parameter tuning, premature convergence, and poor global search performance, which leads to suboptimal controller settings, despite the fact that fractional-order and optimization-based controllers have demonstrated better steady-state and transient behavior. The proposed study combines the Multi-Device Interleaved DC-DC Bidirectional conversion with an intelligent TIFDNFD controller optimized by the Superb Fairy-wren Optimization Algorithm to effectively control the voltage with speed of BLDC motor drivers and reduce overshoot and error. The primary contributions of this manuscript are summarized as follows:

- In order to enhance EV drive performance, a MDIBC is modelled and designed to achieve a steady DC-link voltage across a variety of load circumstances, improved current sharing among interleaved inductors, and decreased input current ripple.
- To control the drives of voltage and speed in the BLDC motor, a new TIFDNFD controller is designed, which is more robust and with a faster dynamic response.
- Through balanced exploration and exploitation, the SFOA algorithm optimizes the controller parameters and speeds up convergence.
- The proposed TIFDNFD-SFOA controller performs better than current control methods Regarding decreased torque ripple, reduced Integral Time Absolute Error (ITAE) and faster settling time, better speed tracking, and minimum overshoot.

The paper is structured as follows: Section 2 presents the literature review, Section 3 describes the System Architecture and Modelling of the proposed MDIBC based EV Drive, Section 4 explains the results and discussion and Section 5 concludes the manuscript.

2. Literature Review

Various research works already exist on converter and motor-drive based EV systems; a brief review of recent studies was presented here.

Anbarasi, M.P., *et al.*, [21] designed, simulated, and analysed of a 10 kW MDIBC, in order to power a 4 kW induction motor. Regardless of changes in the input, a constant 400 V DC output voltage must be maintained. To lower the ripple current in interleaving inductors, MDIBC uses a semi-controlled switch topology that the Phase Shifted PWM method stimulated. Both PI and ANN controllers were used to implement the MDIBC's dual loop

control. The design and control implementations were analyzed and simulated using the MATLAB/Simulink environment.

Kannan, R. and Sundharajan, V., [22] suggested the use of interleaved SEPIC in PEMFCs with MPPT controllers in EVs. As a result, the suggested research was created to create an approach known as MPPT-UFENN. By effectively optimizing the settings, UFENN, one kind on optimized machine learning technique, was developed to optimize the energy extracted from fuel cells. Additionally, In order to maintain the voltage at the output using a minimum of power loss, the interlaced SEPIC conversion was also employed in this work. A comprehensive simulation result was obtained during performance analysis to compare the outcomes of different evaluation indicators with the outcome of the suggested plan.

Dhineshkumar, K., *et al.*, [23] presented a bidirectional DC-DC converter for EV applications that has decreased switch multilayer inverters and integrated MPPT. In order to optimize PV power extraction, this research uses Machine Learning RBFNN MPPT (ML RBFNN MPPT) in conjunction with a SIIFC, or SEPIC Integrative Isolated Flyback Converter. The grid and the EV motor were powered by one VSI and a 31-level inverter with a decreased switch. An adaptive PI controller was used to ensure the EV motor operates at the required speed by synchronizing the grid voltage. The effectiveness of the suggested solar-powered EV charging station was confirmed by experimental validations and MATLAB simulations.

Alzahrani, A., *et al.*, [24] designed and implemented In EV applications, the multi-device interleaved DC-DC boost converter has a higher voltage gain and less voltage stress on the switch. The suggested converter was 3 times more voltage gain compared to a typical Boost DC-DC converter. EVs were among the applications that use these converters because they require a steady DC voltage. The suggested converter's modes of functioning and

mathematical modeling were examined. The viability of the suggested DC-DC converter was confirmed through real-time simulation (OPAL-RT).

Shukla, T. and Kalla, U.K., [25] presented a permanent-magnet the BL-CC converter of a BLDC motor drive for source side PFC in the context of marine EV (MEV) applications. The suggested technique minimizes the number of parts by using a center-tapped inductor in place of the two independent inductors that were normally needed in BL-CC converters. By utilizing the built-in antiparallel diodes of the IGBT switches, the BL-CC converter further simplifies the hardware by incorporating some hardware into other bridgeless topologies that call for additional back-feeding diodes. Although CCM would necessitate multiple sensors, PFC was made possible by the converter's operation in DIC mode, which allows PFC with just one voltage sensor. The architecture also does away with the rectifier stage of the diode bridge. The system was validated under both steady-state and dynamic conditions using MATLAB/Simulink simulations, Bode plots, pole-zero analysis, detailed mathematical modeling, and laboratory experiments.

Huang, C.L., *et al.*, [26] suggested a ripple reduction method of torque using permanent magnetic (PM) BLDC AC motor drives that use the DC-link voltage PAM. The recommended approach lessened the drawback related to the torque ripple of BLDC drives at low speeds. BLDC drives require a low position sensing resolution and have higher inverter efficiency. However, high torque ripples were caused by both discontinuous motor phase current and freewheeling current during diode conduction. This research improves BLDC produces torque ripples by incorporating a DC converter at the front end. While the combination of BLDC drive and DC converter has been documented, this research was the first to regulate the DC voltage used to compensate for BLDC torque ripple. In order to dynamically control the DC current during each BLDC commutation period, When the diode was conducting, the direct current bus was modulated. There was also a sixth-order spatial

harmonic to modulate the DC current in a bid to reduce the impact of the reflected torque ripple as well as consider the discontinuous six-step commutation. Simulation was used to validate the suggested BLDC drive torque ripple compensation for various PM motor types.

Zhou, X., *et al.*, [27] provided an on-board LDC with great power density and efficiency for EV applications. Three- phase interleaved LLC DC-DC converters was paralleled in order to reduce current stress and improve efficiency. Secondary conduction loss can be decreased by using a synchronous rectifier. Switching loss may be significantly reduced through realization of Primary switches were turned on with ZVS, and secondary switches turned off with ZCS. Additionally, light load efficiency was increased through the use of phase-shedding technology. Precise distribution of load current throughout the three phases was accomplished by SCC technology, which lowers conduction loss, safeguards the devices from high-current stress, and increases system reliability. The deterioration in the SCC circuit was less of a problem than the rated output power because SCC switches were made to function on ZVS on/off. Additionally, GaN HEMTs were employed on the main side to increase the density of power and eventually make weight reduction possible.

Kumar, A., *et al.*, [28] commissioned, evaluated, and compared various BLDC motor speed control systems for solar photovoltaic (PV) and grid electricity-powered EVs. The suggested setup makes use of a boost converter and advanced MPPT in a dual-source power system. Nevertheless, feed forward decoupling was effective in controlling power in the grid. Among the motor control techniques that have been evaluated, there were FOC, HCC, and a hybrid approach, including FOC plus HCC (FOC + HCC) and an advanced approach that combines FOC and HCC with a sliding-mode control (FOC + HCC + SMC). This research adds value to the more sustainable and efficient EV technology through providing valuable information on the smart motor control and energy management.

Table 1: Overview of Recent Research Work

Authors	Methodology	Key findings	Advantages	Limitations
Anbarasi, M.P., <i>et al.</i> , [21]	ANN controlled EVs will have a multiple-device interleaved DC-DC boost converter.	ANN enhances transient performance and voltage regulation during load fluctuations.	Reduced input current ripple; improved efficiency	Requires ANN training; increased controller complexity
Kannan, R. and Sundharajan, V., [22]	Interleaved SEPIC converter using MPPT-UFENN	Stable MPPT operation and improved fuel cell utilization	Higher efficiency; reduced voltage stress	Fuel cell cost and degradation not addressed
Dhineshkumar, K., <i>et al.</i> , [23]	Bidirectional DC-DC with ML-RBFNN MPPT+ reduced switch multilevel inverter	High efficiency with regenerative braking support	Reduced switch count and power loss	Complex control strategy; high-power implementation challenges
Alzahrani, A., <i>et al.</i> , [24]	DC-DC boost converter with many devices interleaved	Reduced inductor ripple and improved efficiency	Reduced EMI, experimental feasibility validated	More active components increase overall cost
Shukla, T. and Kalla, U.K.,	BL-CC converter for PFC using	Eliminates separate PFC	Reduced inductor count;	Requires precise

[25]	DIC mode	stage and reduces conduction losses	improved power factor	controller tuning
Huang, C.L., <i>et al.</i> , [26]	DC-link voltage/current modulation for BLDC torque ripple reduction	Significant torque ripple reduction during commutation	Enhanced torque performance and efficiency	Additional modulation and sensing circuitry needed
Zhou, X., <i>et al.</i> , [27]	DC-DC converter with low voltage and great power density.	High-efficiency, compact converter suitable for EVs.	High power density; compact EV converter	Thermal management issues at high power density
Kumar, A., <i>et al.</i> , [28]	FOC + HCC + SMC	Achieves reliable BLDC speed control under hybrid power source	Reduced grid dependency; renewable integration	PV output variation affects performance; MPPT required

2.1 Problem Statement

Recent research on motor drive and converter topologies for EV applications are presented in Table 1. Power conversion and motor drive systems for EVs still face several technical challenges that restrict their effectiveness, dependability, and energy use. In order to maintain a consistent output voltage under a variety of operating situations, existing interleaved DC-DC boost converters need improved control techniques, and traditional PI control might not guarantee optimal performance. Because of the nonlinear properties of fuel cells, it is still challenging to extract fuel cell energy efficiently in EVs; therefore, MPPT accuracy is

essential to achieving maximum available power. Control coordination becomes more challenging when several power converters were integrated with reduced switch multilevel inverters, especially when bidirectional power flow and MPPT are needed at the same time. Conventional boost converters are inadequate for EV systems that require steady high voltage under various input conditions due to their limited voltage gain, increased switching stress, and poor dynamic response. Furthermore, a large number of BLDC drives' current PFC-based converters require additional hardware and a number of inductors. This increases their cost, size, and controller complexity. Significant torque ripple during commutation is another problem for BLDC drives, particularly at low speeds, which impairs efficiency and performance. It is also difficult for present on-board DC-DC converters to achieve high power density and efficiency while maintaining dependable current sharing and reducing switching and conduction losses. Further, fluctuations in solar irradiance and the requirement to coordinate several sophisticated control techniques for enhanced dynamic response face challenges to the optimal speed management of BLDC motors supplied by hybrid sources. To overcome these limitations, the present research discusses a MDIBC combined with a TIFDNFD controller that is optimized using SFOA. The result provides improved voltage and speed regulation, decreased overshoot, and faster dynamic response, making it suitable for advanced EV drive systems.

3. System Architecture and Mathematical Modelling of the Proposed MDIBC based EV Drive

Figure 1 shows the general layout of the proposed BLDC motor drive system based on MDIBC and the TIFDNFD-SFOA controller. As the main power source, the battery provides DC power to the MDIBC, which uses phase-shifted PWM switching and interleaved inductors to increase and control the DC voltage. In addition to improving current sharing

between devices and lowering input current ripple, this interleaving ensures a steady DC link voltage under a range of operating and load scenarios. The Voltage Source Inverter (VSI) transforms the controlled DC link voltage to a three-phase AC supply to power the BLDC motor. By adjusting the pulse width modulation (PWM) generator with response to motor feedback, the TIFDNFD controller precisely controls the BLDC motor's voltage and speed. The SFOA is able to achieve rapid convergence, reduced steady-state error, reduced overshoot, and improved transient performance by tuning the controller parameters. The control loop dynamically modifies the converter duty cycle, continuously monitors motor speed and voltage, and maintains operational efficiency, which makes the system suitable for advanced EV propulsion and industrial motor drive applications.

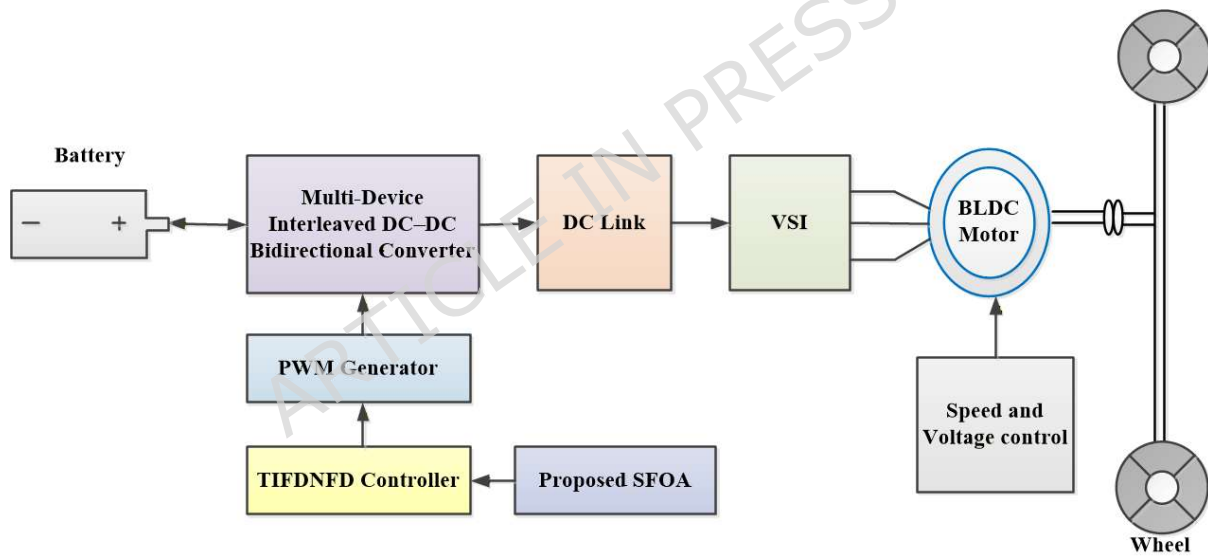


Figure 1: Proposed System Configuration

3.1. Modelling of EV

During motion, EVs must overcome a number of resistive forces, which must be taken into account when dynamically modelling the vehicle [29]. This model makes it possible to determine the energy required to supply the strength and forces of the electric drive in addition to the expected EV load in each drive. As a result, the EVs provide energy and

power according to the vehicle's speed in connection with the driving cycle. However, a number of metrics, such as fuel consumption and pollution levels, are used to evaluate a vehicle's performance. In EV modelling, a variety of roads, velocities, and forces are used. The load is determined by the sum of all forces, which is expressed as:

$$F_L = F_J + F_R + F_f + F_C \quad (1)$$

$$\begin{cases} F_J = 0.5 \cdot \varphi \cdot W \cdot K_U \cdot N_{EV}^2 \\ F_R = (L_{EV} + L_{ESS}) (K_0 + K_1 + L_{EV}^2) \\ F_f = (L_{EV} + L_{ESS}) \cdot \sin(\beta) \cdot f \\ F_C = (L_{EV} + L_{ESS}) \left(u N_{NU} / dt \right) \end{cases} \quad (2)$$

Where L_{ESS} represent the mass of energy storage system, L_{EV} represent the mass of EV, the load force is represented by F_L , F_f represent the gravitational force, F_R represent the rolling resistance, F_C represent the force of acceleration, φ represent the density of electricity, K represent the drag constant, W represent the area, N_{EV} represent the speed of the vehicle and f represent the gravitational constant.

3.2. Modelling of MDIBC

The terms boost converter, interleaved boost converter, and multi-device boost converter are used together as the MDIBC [21] the input inductor is the one which balances the input current. The quantity of converter phases determines the amount of inductors must be utilized. Six semiconductor switches and two parallel devices are used in the conversion. The ratings for both interleaving inductors are the same. The other inductor discharges when one inductor charges at a particular time. The switching frequency is doubled by the inductor current ripple. PWM using phase shifting is used to accomplish this. The total amount of stages and the quantity of simultaneous transistors in each phase define the angle of phase

shift. The MDIBC is characterized by smaller size of the filter, lower ripple factor and low EMI. Since it is limited to low power, the proposed converter that has two input-side filter inductors is highly advised for EVs with low power motors.

3.2.1 Topology of MDIBC

Topology of MDIBC is shown in Figure 2. 400 V DC is added to the 200 V DC input. The input supply is coupled to two two-phase inductors. The inductors L_1 and L_2 have the same values. As shown in Figure 3, the inductor currents are complementary to each other; if the inductor L_1 is charging, the inductor L_2 is discharging. Eight semiconductor switches are in use. The converter has four controlled switches on the lower part of the leg, and four uncontrolled diodes on the upper part. MOSFET switches are used with switching pulses of 20 kHz frequency. Equation (3) provides the relationship between the converter's input and output voltages.

$$Q = \frac{1}{n} \left(1 - \left(\frac{V_{dc}}{V_o} \right) \right) \quad (3)$$

Where, Q represent the duty cycle, n represent the number of phase, V_{dc} represent the input voltage and V_o represent the output voltage. Equation (4) can be used to determine the values of the inductors used in the front as boost inductors because they are all rated the same.

$$L = \frac{(1 - nP) \times V_o \times Q}{d \times F_{sw} \times \Delta I_{L_{max}}} \quad (4)$$

Where, L represent the filter inductor, F_{sw} represent the switching frequency, d represent the number of parallel path and $\Delta I_{L_{max}}$ represent the maximum inductor current. Equation (5) expresses the capacitor's value.

$$C = \frac{V_o \times Q}{d \times F_{sw} \times R \times \Delta V_o} \quad (5)$$

Where, C represent the capacitor value and R represent the resistance.

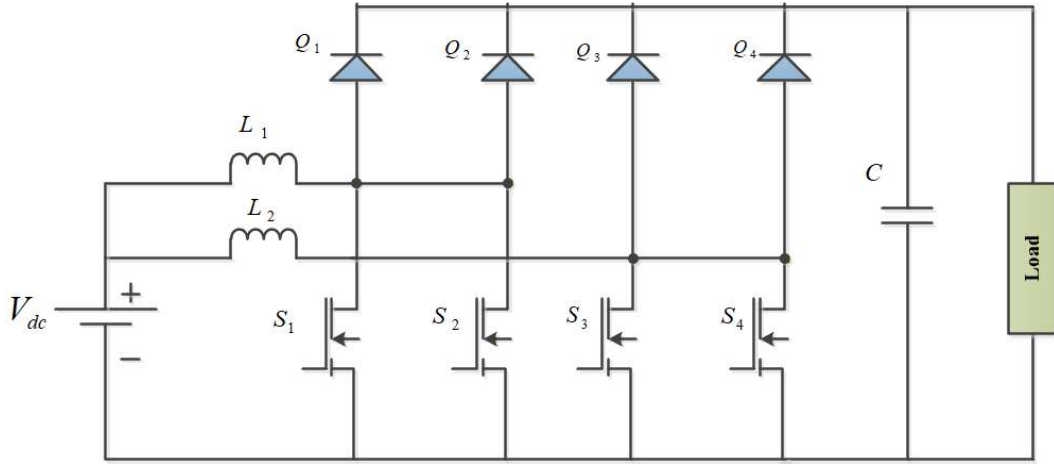


Figure 2: Topology of MDIBC

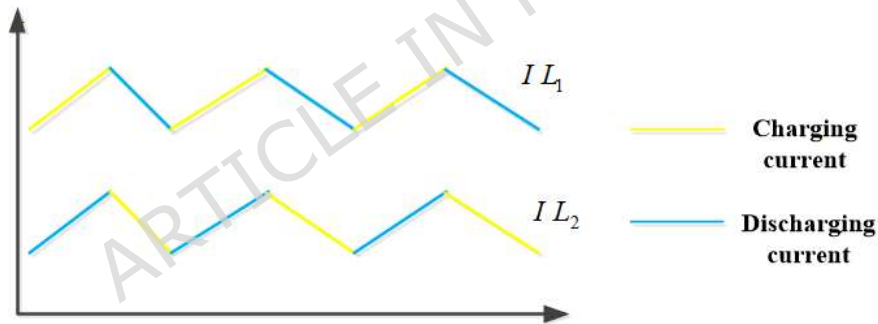
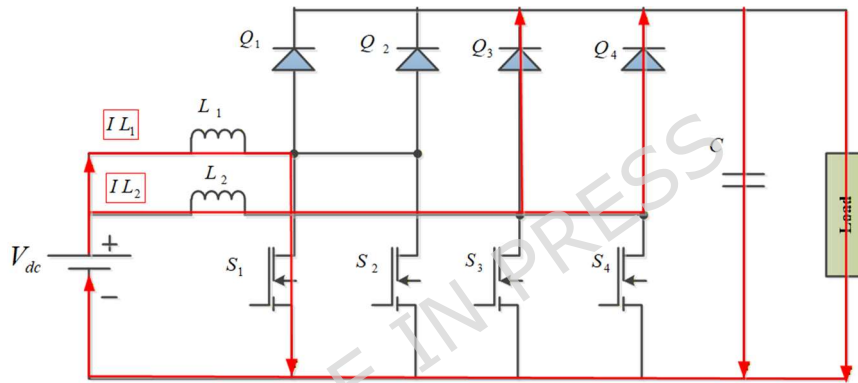


Figure 3: Inductor Currents

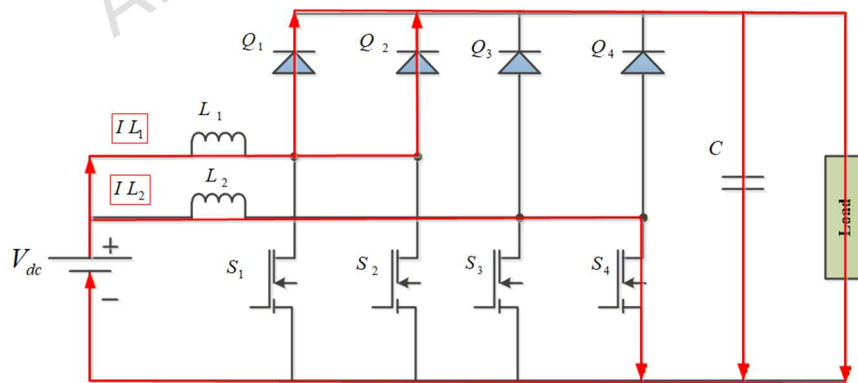
3.2.2 MDIBC Operating in Boost Mode

Medium power electric motors power daily commuting EVs. The crucial part for driving the motor is the power electronic interface unit. The battery, an EV's primary power source, must transform DC voltage into AC in order to operate the traction motor. The DC-DC converter would supply ripple-free DC to the inverter in place of the battery. It is essential to have a

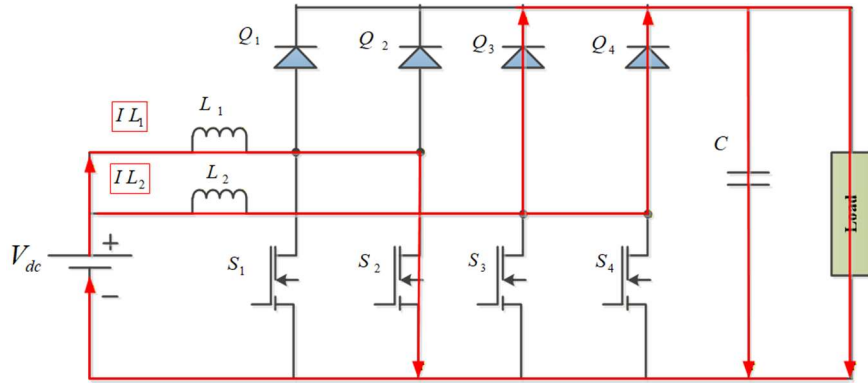
converter that converts DC to DC, is dependable, has a lower filter size, low voltage and current ripple, and improves battery performance quickly. DC-DC converters should use lightweight, compact filters. To meet the requirement, a multi-device intermittent boost converter is selected. Its primary function is to raise the battery's voltage. A converter combines interleaved boost, boost, and multi-device converters. Only the upper and lower leg switches are activated during the boost operating mode. MDIBC contains two current smoothing inductors, one for each phase, in accordance with the boost converter's architecture. The output DC voltage ripple is reduced by the capacitor placed across the load.



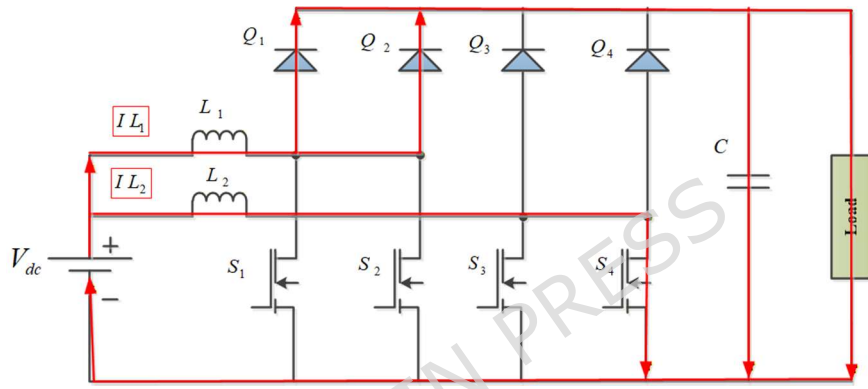
(a)



(b)



(c)



(d)

Figure 4: Current flow when (a) switch S_1 is ON at $\delta = 0^\circ$.

(b) $\delta = 90^\circ$ when switch S_3 is ON. (c) $\delta = 180^\circ$ when switch S_2 is ON. (d) switch S_4 is ON at

$$\delta = 270^\circ$$

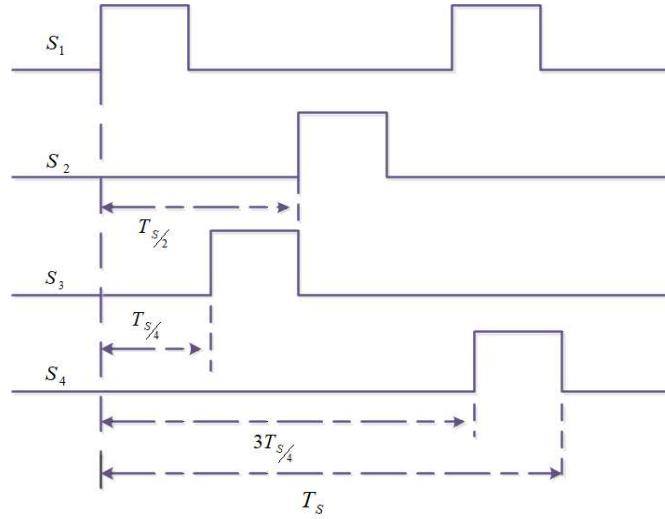


Figure 5: PWM Sequence Phase shift

Figure 4 shows the converter's switching sequence. Figure 5 displays the switch time period PWM pattern determines the delay times of four pulses. Furthermore, Figures 4 and 5 show the switching order, current flow, and pulse production of the devices. Current $I L_1$ flows through S_1 and $I L_2$ flows through $Q_3 - Q_4$ and the load when S_1 is turned ON ($\delta = 0^\circ$, time = 0 s). When S_3 is turned on ($\delta = 90^\circ$, time = $T_{s/4}$), $I L_1$ returns through $Q_1 - Q_2$ and the load, while $I L_2$ flows through S_3 . Current passes through S_2 , the complementary path through $Q_3 - Q_4$, and the load when S_2 turns on ($\delta = 180^\circ$, time = $T_{s/2}$). Finally, $I L_1$ returns to the source and $I L_2$ passes through $Q_1 - Q_2$ and the load when S_4 is turned ON ($\delta = 270^\circ$, time = $3T_{s/4}$). The PWM generator is used to generate four PWM pulses. For every switch, the duty cycle is the same. The PWM phase variation for the switching pattern of the eight switches is found using equation (6). One way to calculate the phase shift angle is:

$$\delta = \frac{360^\circ}{(n \times d)} \quad (6)$$

Where, δ represent the angle of phase shift, n represent each channel's parallel switches and d represent the quantity of channels or phases. This phase-shifted operation guarantees minimal ripple and balanced current sharing.

3.3 Modelling of BLDC Motor

The equivalent circuit of a BLDC motor, as depicted in Figure 6 [31], can serve as a mathematical model of the motor's configuration. The resistance (R_s) and corresponding inductance for each phase (L_s) indicate the stator windings. The return electromotive force (EMF) each phase of a BLDC motor exhibits a trapezoidal waveform rather than the typical sinusoidal shape. There is a 120 electrical degree gap between the back-EMFs. Its three-phase inverter that enables its motor is capable of detecting the rotor magnetic field to switch at the right time due to three Hall sensors fitted in the stator frame of the three phase VSI and assuming full-pitch windings and a star-connected stator, the BLDC motor's phase voltage can be written as follows:

$$V_a = I_a R_a + (L - M) \frac{di_a}{dt} + e_a \quad (7)$$

$$V_b = I_b R_b + (L - M) \frac{di_b}{dt} + e_b \quad (8)$$

$$V_c = I_c R_c + (L - M) \frac{di_c}{dt} + e_c \quad (9)$$

Where phase voltages are represented by the variables V_a , V_b , and V_c ; the representation of armature resistance by R_a , R_b , and R_c ; L represents the armature self-inductance; M represents the mutual inductance; I_a , I_b , and I_c represents the phase currents; e_a , e_b , and

e_c represents the motor back emf of phases "a", "b", and "c" respectively. Where the back emf of the BLDC motor can be described as follows:

$$e_a = NS \frac{d}{dt} \left[\int_{-\frac{\pi}{2} + \theta_e}^{\frac{\pi}{2} + \theta_e} B(\Theta) d\Theta \right] = 2NS\omega B \left(\frac{\pi}{2} + \theta_e \right) \quad (10)$$

In the air gap, the back emf wave is $\frac{\pi}{2}$ ahead of the magnetic density distribution and depends on θ_e . Therefore, e_a can be expressed as follows:

$$e_a = \underbrace{2NSB_m}_{K_b} \omega \lambda_a(\theta_e) = K_b \omega \lambda_a(\theta_e) \quad (11)$$

where N represent the number of turns per pole, S represent the product of the rotor radius and the effective length of the conductor, B_m represent the maximum value of the flux density in the air gap and ω represent the angular speed of the motor respectively. The back emf waveform is represented by the $\lambda_a(\theta_e)$, whose value varies with the distribution of air gaps. Likewise, the back emf of phases "b" and "c" can be expressed as follows:

$$e_b = K_b \omega \lambda_a \left(\theta_e = \frac{2\pi}{3} \right) = K_b \omega \lambda_b(\theta_e) \quad (12)$$

$$e_c = K_b \omega \lambda_a \left(\theta_e = \frac{4\pi}{3} \right) = K_b \omega \lambda_c(\theta_e) \quad (13)$$

As long as the motor is running, if the disregard the losses of iron and copper, the overall amount of power taken in from the source is sent to the rotor via the air gap, which is referred to as electromechanical power and expressed as follows:

$$P_e = P[K_b \omega \lambda_a(\theta_e) I_a + K_b \omega \lambda_b(\theta_e) I_b + K_b \omega \lambda_c(\theta_e) I_c] \quad (14)$$

Where, P represent the quantity of pole pairs. Additionally, the entire electromechanical power is transformed into kinetic energy, which can be represented as electromagnetic torque in the following equation, ignoring mechanical and stray losses:

$$T_e = \frac{P[K_b \omega \lambda_a(\theta_e)I_a + K_b \omega \lambda_b(\theta_e)I_b + K_b \omega \lambda_c(\theta_e)I_c]}{\omega} \quad (15)$$

Additionally, the torque calculation equation can be expressed like this in terms of back emf:

$$T_e = \frac{1}{\omega} \sum_{k=a,b,c} e_k I_k \quad (16)$$

The mathematical expression for converting electromagnetic torque into mechanical motion is expressed as:

$$T_e = G_m \frac{d\omega}{dt} + A_v \omega + T_L \quad (17)$$

Where, T_L represent the torque of the load, G_m represent the inertia moment and A_v represent the viscous friction coefficient, respectively.

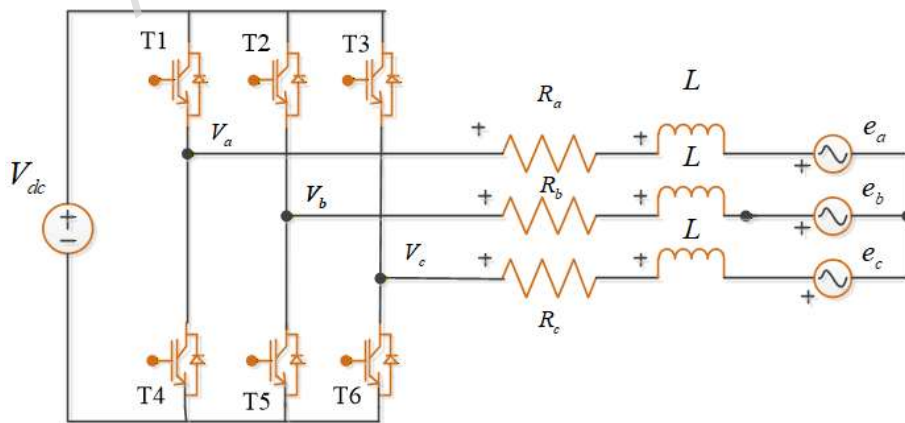


Figure 6: Equivalent BLDC motor circuit

3.4. Modeling of TIFDNFD Controller

The TIFDNF controller is an amalgamation between the FOPID and TID controller [30]. The TID controller takes the derivative component and changes it to a fractional component after which the integer component of the order driven filter which includes the additional FO derivative term is added and the resulting controller is the TIFDNFD controller. The fractional version of the integral is not used here. This is due to an increase in the frequency oscillation and tie-line power of the simulations as indicated by the FO-based integral. In general, the controller's fine-tuning capability increases with the number of controller setting parameters, which enhances the controller's control performance. With its eight tuning parameters, TIFDNFD aims to achieve this. The TIFDNFD controller is expressed as

$$C_{TIFDNFD}(s) = K_t \frac{1}{s^n} + K_i \frac{1}{s} + K_d s^{\mu_1} \left(\frac{N}{N+s} \right) K_{dd} s^{\mu_2} \quad (18)$$

$$u(t)_{TIFDNFD} = \left(K_t \frac{1}{s^n} + K_i \frac{1}{s} + K_d s^{\mu_1} \left(\frac{N}{N+s} \right) K_{dd} s^{\mu_2} \right) e(t) \quad (19)$$

$$u(t)_{PID} = \left(K_p + K_i \frac{1}{s} + K_d s \right) e(t) \quad (20)$$

$$u(t)_{TID} = \left(K_t \frac{1}{s^n} + K_i \frac{1}{s} + K_d s \right) e(t) \quad (21)$$

$$u(t)_{FOPID} = \left(K_p + K_i \frac{1}{s^\lambda} + K_d s^\mu \right) e(t) \quad (22)$$

Where, μ_2 , K_t , K_{dd} , μ_1 , K_d , n , K_i , and N represent the tuning parameters of *TIFDNFD* controller. λ represent the integral order. Figure 7 shows the TIFDNFD controller block diagram,

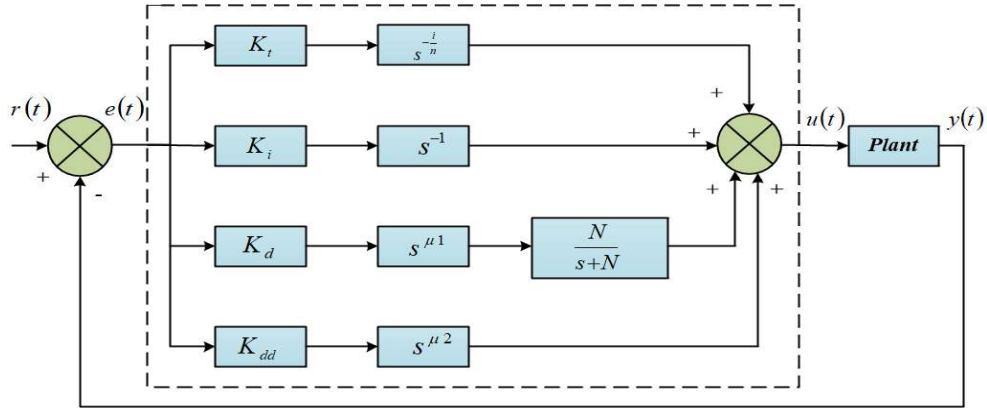


Figure 7: TIFDNFD Controller Block Diagram

3.5. Controller Parameter Tuning Using the Superb Fairy-wren Optimization Algorithm

In this section, TIFDNFD controller parameters (K_t , K_i , K_{dd} , K_d , n , μ_1 , μ_2 and N) are optimally tuned using the proposed SFOA algorithm [32]. The design of the proposed SFOA method modifies the population member's position in the problem-solving space by mimicking the natural behavior of the lovely fairy-wren juveniles and adults. The SFOA algorithm's concept originates from the cooperative breeding and communication behavior of the superb fairy-wrens. In this natural process, the young bird first goes through the growth stage, when it can either use the defence system to fend off predators or the feeding code to hatch the young when it reaches the maturity stage. Thus, each time an assessment is conducted, the status of SFOA population members will be updated in three stages: (i) intensive research grounded on the development of young birds, (ii) exploitation through nourishment code breeding, and (iii) exploitation through the predator defence behavior simulation of wrens. The proposed SFOA algorithm's mathematical formulation is provided below.

Step 1: Initialization

The presented SFOA approach is a population approach that effectively tackles the real environment's optimization challenge by mimicking its spatial search capabilities by altering evaluation durations. Each SFOA member determines the value of the problem decision variable based on the problem's article inside the search space. Each SFOA member is a potential solution to the issue, and they are all represented mathematically as vectors. Each vector element in the algorithm, which consists of all SFOA members, represents a decision variable that is represented in Equation (23).

$$Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_2 \\ \vdots \\ Y_N \end{bmatrix}_{N \times D} = \begin{bmatrix} y_{1,1} & \cdots & y_{1,d} & \cdots & y_{1,D} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ y_{i,1} & \cdots & y_{i,d} & \cdots & y_{i,D} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ y_{N,1} & \cdots & y_{N,d} & \cdots & y_{N,D} \end{bmatrix}_{N,D} \quad (23)$$

Therefore, Equation (24) is used to initialize each SFOA member at the algorithm's first main site.

$$Y = (UB - LB) \times rand(0,1) + LB \quad (24)$$

Where, Y represent the SFOA global matrix, y_i represent the i^{th} SFOA member (possible solution), N represent the number of global members, $rand$ represent the random number in the interval $[0,1]$, and $y_{i,D}$ represent the D dimensions of SFOA in the search space (decision variables). The decision variables' upper and lower bounds are represented by UB and LB respectively.

Step 2: Fitness Function

The multi-objective function is utilized to calculate the fitness of each agent:

$$Fitness\ Function = Min\ (ITAE) \quad (25)$$

Where $Min (ITAE)$ represent the minimizing integral of time-weighted absolute error.

Step 3: Young bird's stage of growth

They use dynamic simulations, which require a great deal of information to produce young birds, to update the population's placements during the SFOA growth phase. Since the enormous initial of young birds in the population is detrimental to the population's survival, SFOA members duplicate their position in the problem-solving space by continuously collecting an enormous amount of expertise with updating it as young birds develop swiftly. Their process motions, which are referred to as the learning-from-experience will aim at substantially changing the positioning of the SFOA members, therefore, enhancing the degree of exploration offered by the global search algorithm. In order to enhance the objective response SFOA employs Equation (26) to establish the position of every member.

$$Y_{new_{i,j}} = Y_{i,j}^{ite} (LB(UB - LB)) \times rand, \quad r > 0.5 \quad (26)$$

Where, $Y_{new_{i,j}}$ denotes the position following population renewal, and $Y_{i,j}$ denotes the position of the i^{th} superb fairy wren in j dimension following ite iterations.

Step 4: Feeding and Breeding Stage

In the second step of SFOA, the second generation updated the positions of the members of the population by imitating the process of nursing and breeding of the superb fairy-wren. The wrens utilize unique paternity tests when incubating their eggs and reproduce at low risk levels to prevent the spread of foreign species. Equation (27) illustrates the danger threshold S calculation process.

$$s = r1 * 20 + r2 * 20 \quad (27)$$

Where, $r1$ and $r2$ represents the random numbers have a normal distribution. Multiple birds incubate eggs throughout the year to achieve the teaching of identification because of the cooperative reproduction characteristics of SFOA. Each SFOA moves around during this cycle (w), taking turns feeding and instructing. SFOA members positions slightly alter when this phenomenon is modelled, which increases the algorithm's use for local searches. Meanwhile, establish a factor q that develops as the length of the teaching cycle shortens. Each SFOA member engages in a wider range of activities as they get closer to maturity. Equation (28) was used to establish each SFOA member's new position after the teaching eggs' location movement during breeding was simulated. The matching member is then substituted if the objective function's value increases as a result of this updated position.

$$Y_{new_{i,j}} = Y_G (Y_b Y_{i,j}^{ite}) \times q, \quad r < 0.5 \text{ and } s < 20 \quad (28)$$

$$Y_G = Y_b \times C \quad (29)$$

Where, C represent the constant and Y_b is the current ideal position.

$$q = \sin((UB - LB) \times 2 + (UB - LB) \times w) \quad (30)$$

$$w = \left(\frac{E}{Max E} \right) \times 2 \quad (31)$$

Where, $Max E$ represent the maximum number of evaluation and E represent the current number of evaluation.

Step 5: Avoiding the stage of natural enemies

In SFOA's predator avoidance phase, the population's relative positions are also changed based on the magnificent fairywren's defence mechanism against predator assault. When the

predator spots the amazing fairy-wren, it flees while flapping its wings to avoid being seen. Additionally, to alert the other SFOA members, the alert sound is played. An intended target SFOA member flees swiftly in such a scenario, slightly altering the member's location. When they flutter in the air to evade the predators, the other individuals' posture radically alters. Equation (32) shows that an algorithm's search range in the problem-solving space and its capacity for local searches are expanded by the two distinct expressive pressures of the SFOA members.

$$Y_{new_{i,j}} = Y_b + Y_{i,j} \times l \times k, \quad r < 0.5 \text{ and } s > 20 \quad (32)$$

The control algorithm departs from the local optimal when l represent the Levy flight random step size. Equation (33), which controls the birds' flight distance in conjunction with l , uses k , or the adaptive flight balance factor. Simultaneously, Y_b was added to regulate the birds' movement direction in order to stop them from heading in a direction of the unfavourable position, which would have wasted evaluation time.

$$k = 0.2 \times \sin\left(\frac{\pi}{2} - m\right) \quad (33)$$

$$m = \frac{\pi}{2} \times \frac{E}{Max E} \quad (34)$$

Where, m represent the value of the call frequency, it acts as a warning mechanism to steer clear of natural adversaries while in flight.

Step 6: Termination

The optimization process continues iteratively until a stopping criterion is satisfied. Termination occurs when the maximum number of function evaluations ($Max E$) is reached.

Otherwise, follow the previous steps and continue updating the population until convergence.

The SFOA algorithm's pseudocode is shown in Algorithm 1.

Therefore, the proposed SFOA algorithm provides better tuning of TIFDNFD controller parameters and increased efficiency.

Algorithm 1: Pseudocode of SFOA Algorithm

Initialize the population and parameters
Determine the initialized population's fitness and identify the most optimal individual
While $E \leq \text{Max } E$
For $i = 1$ to N
If $r > T$ (A higher percentage of the population consists of young birds.)
To improve the objective function, SFOA uses Equation (26) to determine each member's position.
Else
Determine the current position's danger factors S using equation (27)
If $S < r$ (Feeding and breeding stage)
The breeding process was conducted using equation (28)
Else (Avoiding natural enemies stage)
Avoiding the natural enemies using equation (32)
End If
End If
End For

End While
Calculate the best individual by calculating the updated population's fitness ($E = E + 1$)
End

4. Results and Discussions

To implement the proposed approach, MATLAB/SIMULINK is used and its efficacy is confirmed. The proposed TIFDNFD-SFOA controller effectively ensures superior transient response, minimal overshoot and reduced error in EV systems. The Proposed TIFDNFD-SFOA controller was implemented and outperforms traditional FOSMC [33], SIIFC-RBFNN [23] and TIFDNFD-based controllers to demonstrate its effectiveness in terms of the minimum overshoot, peak, and settling times. Figure 8 shows the Simulink model.

Table 2: Simulation Parameters

Parameter	Value
iteration	100
population	20
dim	5
LB	-10
UB	10
TIFDNFD controller	
K_t	1.46

K_i	12.8
K_d	0.00018
K_{dd}	0.0054
n	3
μ_1	0.5
μ_2	0.5
λ	0.5
N	5
Vin	250V
Inverter	
R_s	5000 ohm
C_s	1e-6 F
BLDC Motor	
Stator phase resistance R_s (Ohm)	2.8750
Stator phase inductance L_s (H)	8.5e-3
Back EMF flat area (degrees)	120

Table 2 shows simulation parameters. To evaluate the intrinsic dynamic performance of the proposed MDIBC and TIFDNFD–SFOA controller, a light-load operating condition of the inverter–DC link stage is represented by the load resistance value ($R_s = 5000 \Omega$). Reduced load current under these circumstances minimizes excessive ripple and demonstrates how well the controller regulates voltage and speed. Similar steady behaviour was demonstrated

by additional test with lower resistance values, indicating that the selected resistance is appropriate for verifying the control performance and system dynamics.

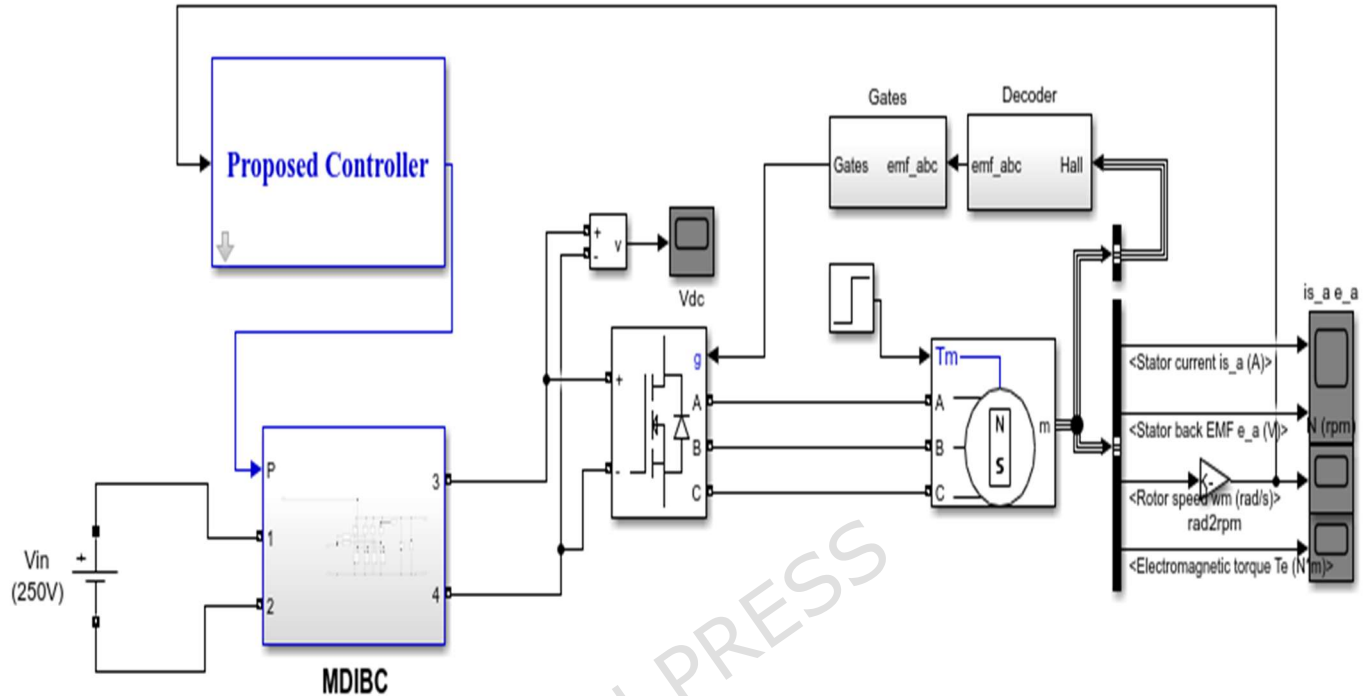
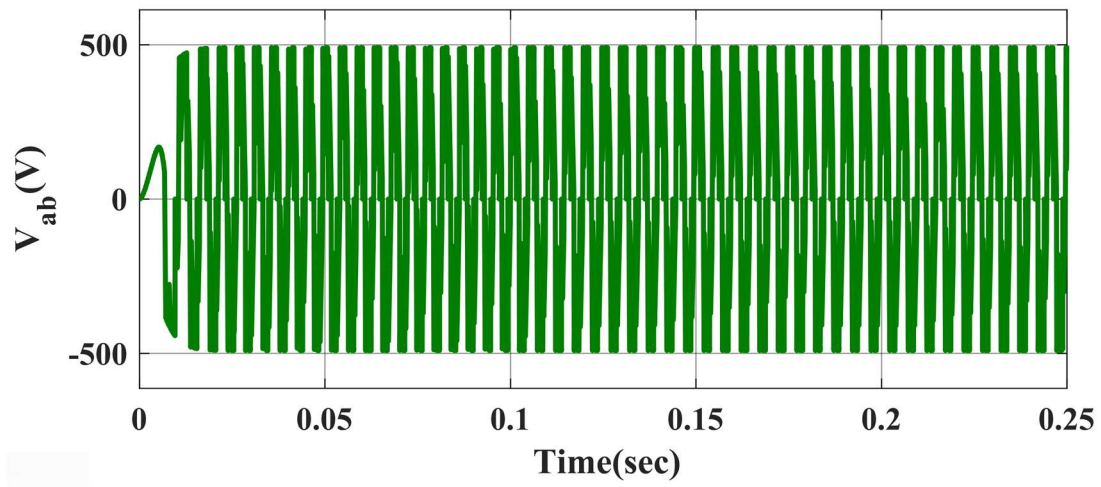


Figure 8: Simulink Model

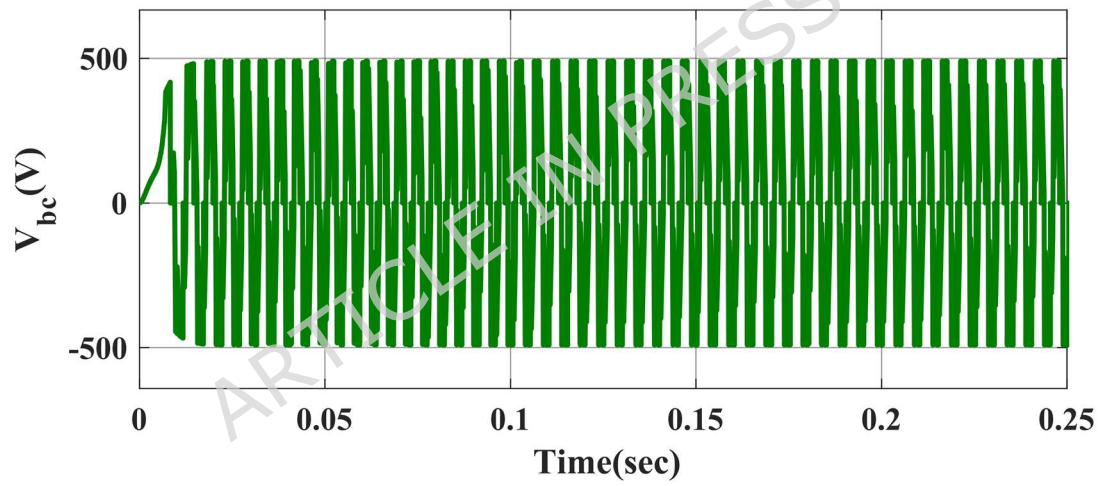
4.1 Analysis of Three-Phase Output Voltage of Voltage Source Inverter

The three-phase line-to-line output voltages produced by the Voltage Source Inverter (VSI) are displayed in Figure 9. Figure 9 (a) shows a quick transient response, with the voltage V_{ab} reaching a peak amplitude of roughly +500 V to −500 V and the waveform stabilizing after 0.01 sec. Figure 9 (b) displays V_{bc} , which is phase-shifted by 120 electrical degrees in relation to V_{ab} and also ranges between ± 500 V, indicating correct inverter switching synchronization. Figure 9 (c) completes the balanced three-phase sequence by displaying V_{ca} with an extra 120° phase shift and the same peak magnitude (± 500 V). The voltages stay balanced around 0 V in all three figures, and the consistent sinusoidal pattern shows that the

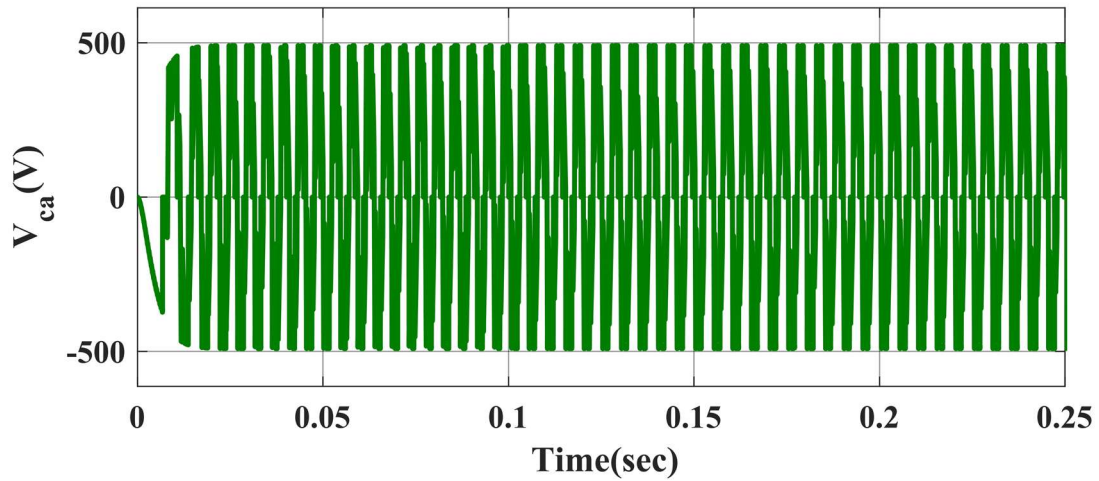
inverter effectively transforms the boosted DC of the MDIBC to a three phase AC supply which is symmetrical and suitable to drive BLDC motors.



(a)



(b)



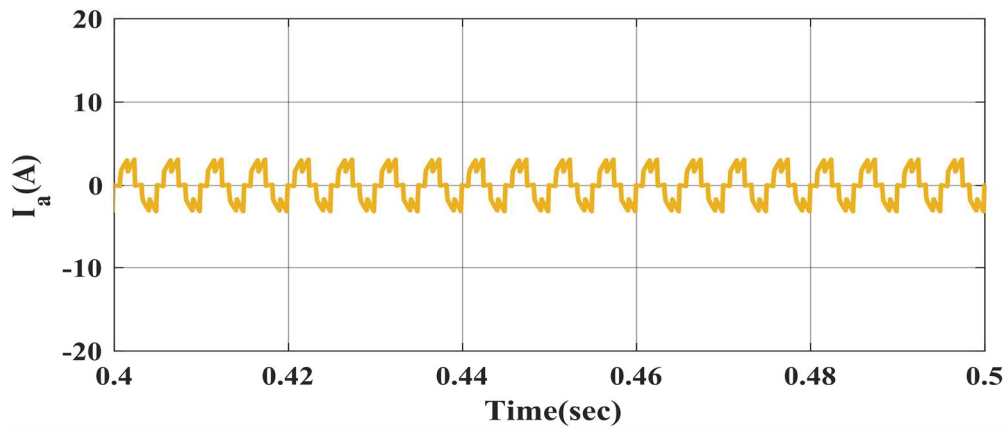
(c)

Figure 9: Three-Phase Output Voltage of Voltage Source Inverter

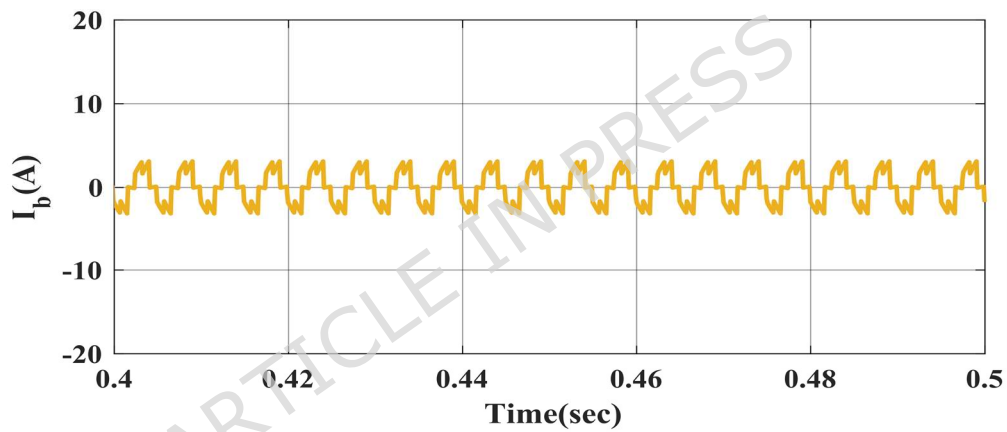
4.2 Three-Phase Stator Current Analysis

The BLDC motor's simulated three-phase stator currents for phases are shown in Figure 10. The phase current I_a in Figure 10 (a) oscillates between roughly +8 A and -8 A and stabilizes within the time span of 0.40–0.50 s, suggesting that the controller controls the phase current symmetrically around zero. Figure 10 (b) displays the phase current I_b , which also maintains a peak current value of ± 8 A and shows the anticipated electrical phase shift of 120° in relation to I_a , indicating correct commutation. The balanced three-phase current sequence is completed by Figure 10 (c), which displays the current I_c , which is likewise centered around zero with an amplitude that varies between +8 A and -8 A and moved 120° behind I_b . The proposed MDIBC and TIFDNFD-SFOA controller maintains steady and symmetrical current injection into the motor stator windings, as shown by the

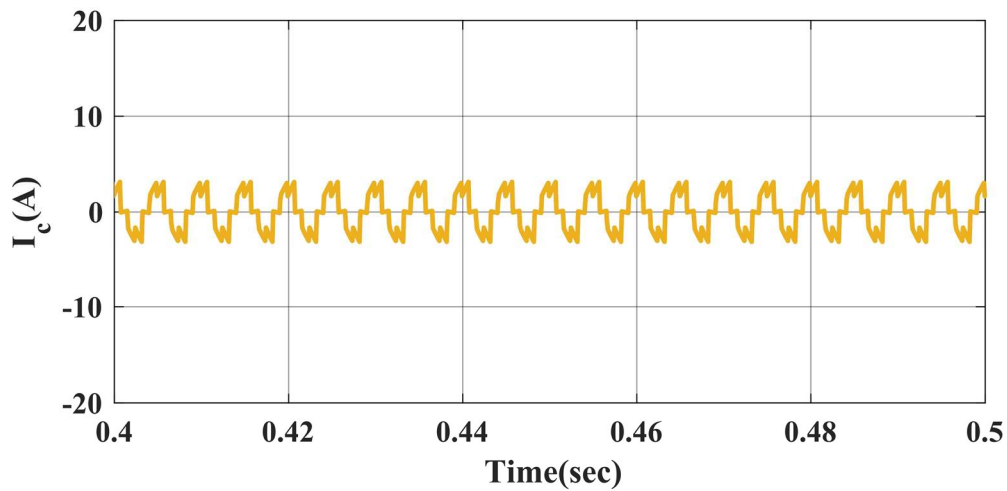
currents constant size and periodic ripple caused by PWM switching across all three waveforms.



(a)



(b)

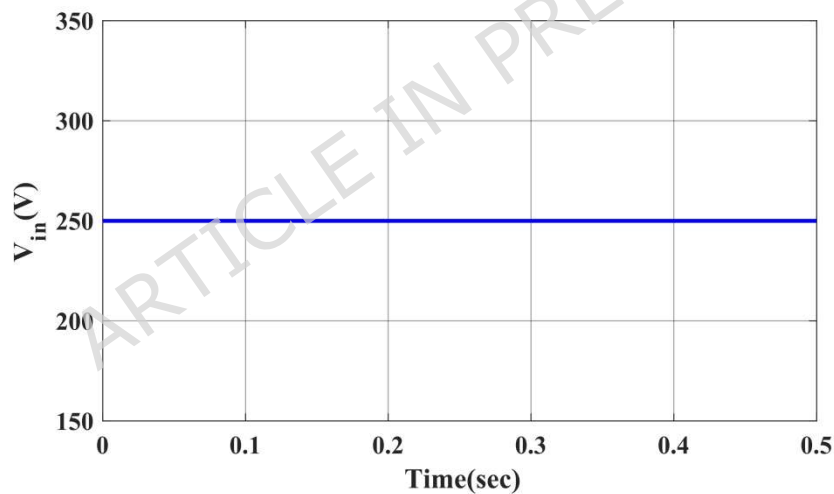


(c)

Figure 10: Simulated Three-Phase Stator Current

4.3 Input Voltage Analysis

The input voltage applied to the MDIBC during the simulation period, which lasted from 0 to 0.5 sec, is depicted in Figure 11. The plot clearly shows that there are no variations or disruptions to the input voltage, which stays steady at 250 V. A steady DC supply to the converter is ensured by the flat horizontal line, which shows that the input voltage remains constant at 250 V even under dynamic operating conditions of the converter and BLDC motor load variations. This demonstrates that the input to the MDIBC is regulated and free of ripple, which is necessary to preserve steady boost converter performance and avoid putting the inverter stage under undue voltage stress.

**Figure 11:** Analysis of Input Voltage

4.4 Output Voltage Analysis

The output voltage of the MDIBC during the 0–0.5 sec simulation interval is shown in Figure 12. The output voltage quickly rises from 0 V at start up and reaches about 480 V in the first 0.02 sec, indicating a quick dynamic response and efficient boost action of the converter.

Once this level is reached, the voltage stays constant at about 480 V for the rest of the simulation without any discernible ripple or oscillation, demonstrating that the proposed MDIBC and controller continue to produce a stable boosted output. The BLDC motor can operate effectively and without ripple due to the steady output voltage, which guarantees a steady DC link supply to the inverter.

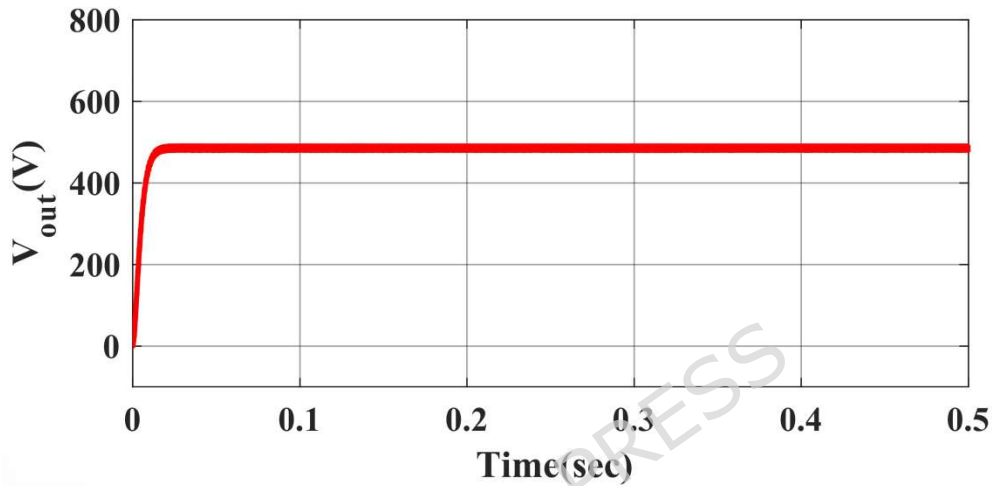


Figure 12: Analysis of Output Voltage

4.5 Back Electromotive Force (emf) of the BLDC Motor Analysis

Figure 13 shows the trapezoid back emf waveform of phase-a of the back emf of the BLDC motor in the steady operating area of 0.40 0.50 sec. The BLDC motor is confirmed to be running at its rated speed by the waveform's consistent alternating between roughly +220 V and -220 V, resulting in a peak-to-peak magnitude of about 440 V. The waveform's flat tops and sudden changes show that the motor is effectively commuting with the inverter switching sequence and generating the anticipated trapezoidal back-emf profile. Besides establishing that the offered MDIBC and controller transmit a fixed DC link voltage, the constant amplitude throughout the time period indicates that the motor is running consistently, which

guarantees that torque production is always smooth and that the BLDC has a proper functioning.

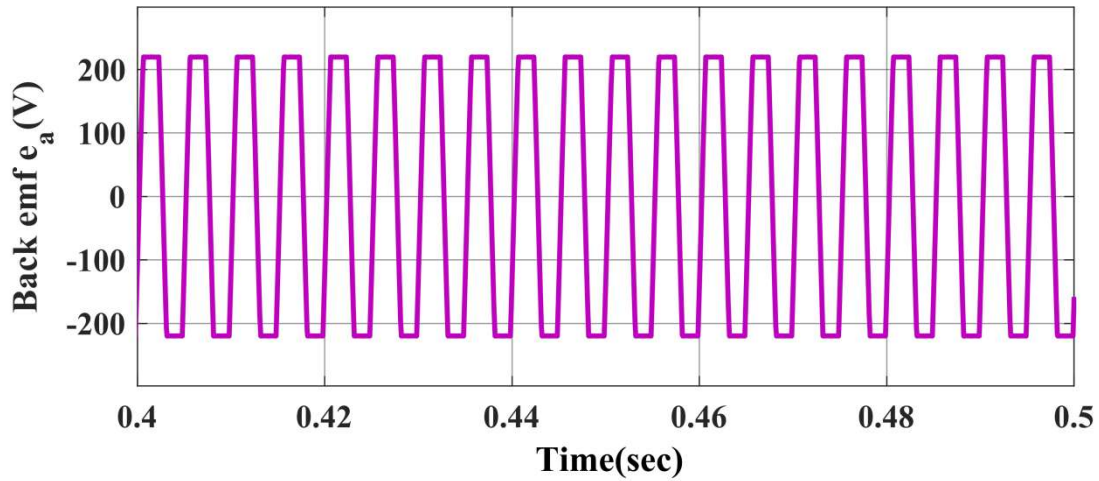


Figure 13: The back emf of the BLDC motor

4.6 BLDC Motor Speed Analysis

The BLDC motor's speed response from 0 to 0.5 seconds during the simulation is shown in Figure 14. The motor has a quick transient response and efficiently produces torque at beginning, accelerating from 0 rpm to about 3200 rpm in the first 0.05 seconds. The motor rapidly settles to a steady operating speed of roughly 3000 rpm after a slight overshoot at 3300 rpm that occurs before 0.10 seconds. After 0.10 seconds, there are no discernible oscillations or fluctuations in the speed at 3000 rpm, suggesting that the proposed MDIBC combined with the TIFDNFD-SFOA controller offers stable operation under load conditions, high speed regulation, and low error.

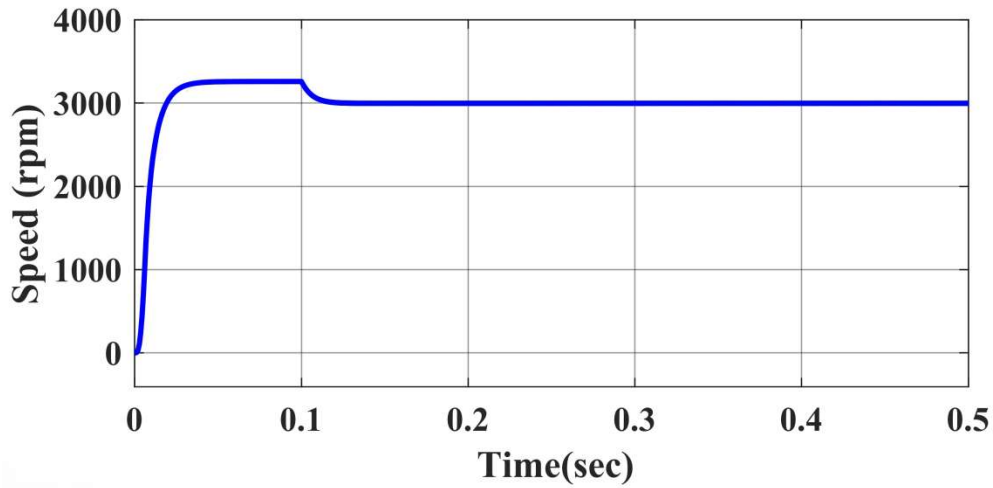


Figure 14: Analysis of BLDC Motor Speed

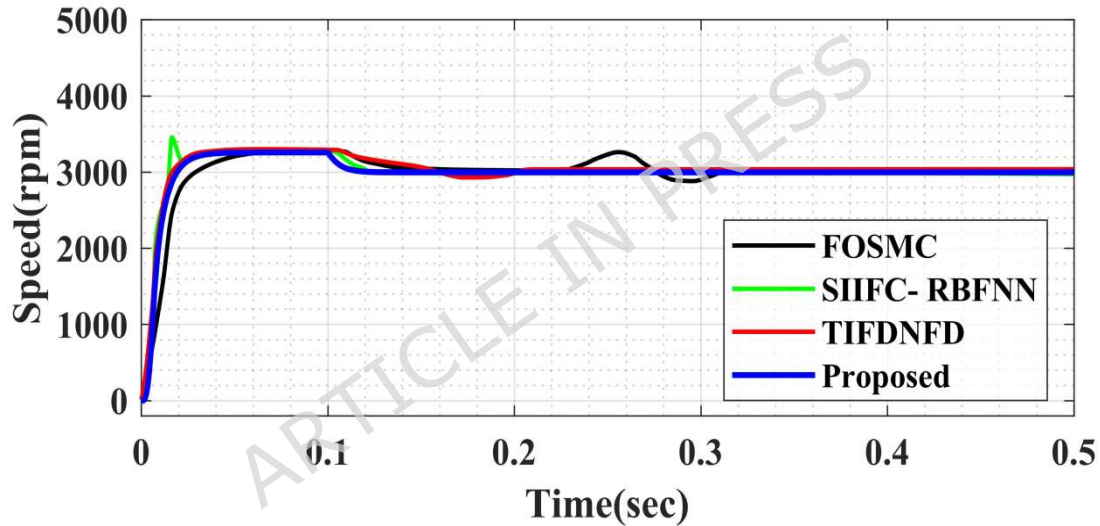


Figure 15: BLDC motor speed response comparison using proposed and existing controllers.

Figure 15 illustrates a case of the comparison between the response of the BLDC speed with four different controllers, including: FOSMC, SIIFC-RBFNN, TIFDNFD, and the proposed controller, TIFDNFD-SFOA, to a simulation period of 0 to 0.5 seconds. The motor is accelerated by all controllers from 0 rpm to the rated speed of 3000 rpm, but there are notable differences in their dynamic behaviours. A moderate overshoot of about 3300 rpm is displayed by the TIFDNFD controller, whereas the SIIFC-RBFNN controller shows the

largest overshoot, reaching about 3500 rpm before settling. A minimum peak time and significant oscillations between 3000 and 3200 rpm, particularly around 0.25 seconds, are produced by the FOSMC controller, suggesting worse steady-state performance and speed regulation. The proposed TIFDNFD-SFOA controller performs better than the existing controllers. It enables the motor to take approximately 3100 rpm with a small overshoot with a steady-state velocity of 3000 rpm within approximately 0.12 seconds. The flat and ripple-free response following settling demonstrates that the proposed controller provides the quickest settling time, the least amount of speed fluctuation under load, and superior transient performance.

4.7 Analysis of BLDC Motor Torque

Figure 16 indicates the electromagnetic torque response of the BLDC motor in the 0-0.5 seconds simulation period. To overcome the motor's inertia and start acceleration, a high initial torque peak of about 38 Nm is produced at beginning. Within the first 0.05 seconds following this peak, the torque rapidly drops and settles close to 0 Nm, demonstrating that the motor achieves the desired speed with little oscillation. As the motor enters steady-state operation, a slight torque demand of about 5 Nm appears at 0.10 seconds. The torque then steadily and smoothly stabilizes for the duration of the simulation. The fact that the proposed MDIBC and TIFDNFD-SFOA controller is capable of minimizing torque ripple and providing a smooth and efficient operation of the motor is attested by the absence of ripples and oscillations subsequent to settling.

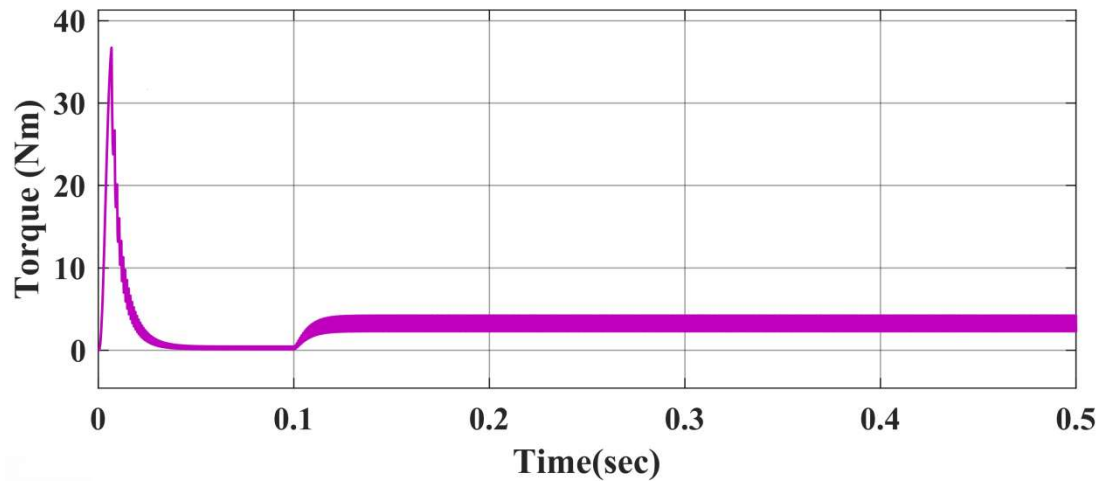


Figure 16: Analysis of BLDC Motor Torque

4.8 Analysis of ITAE

Figure 17 shows how the ITAE varies between 0 and 1 second for both proposed and current controllers. The largest error magnitude is shown by the FOSMC controller, which reaches about 0.10, then the SIIFC-RBFNN, which peaks at about 0.08, and the TIFDNFD controller, which reaches about 0.05. The proposed TIFDNFD-SFOA controller exhibits superior transient tracking accuracy with a substantially lower initial ITAE value that stays near 0.02. While all controllers gradually approach 0 ITAE, the proposed controller reaches 0 in less than 0.02 seconds. The proposed controller offers the most effective error minimization, as shown by the faster decay and smaller error magnitude, which enhances dynamic performance and lowers steady-state deviation.

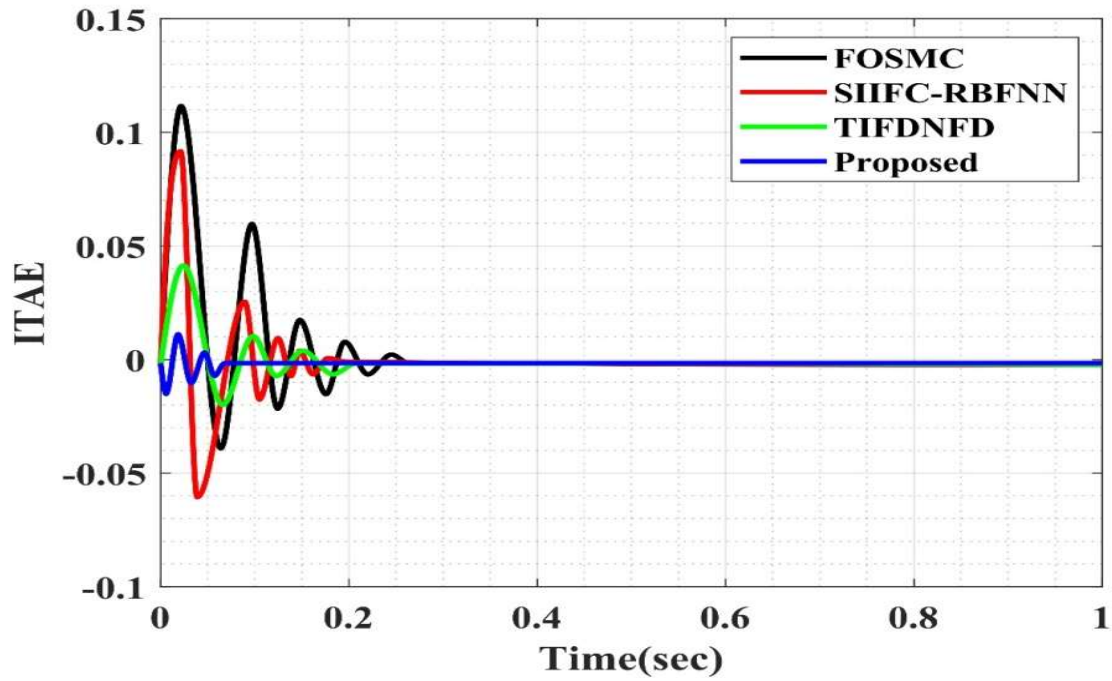


Figure 17: Analysis of ITAE

4.9 Convergence Analysis

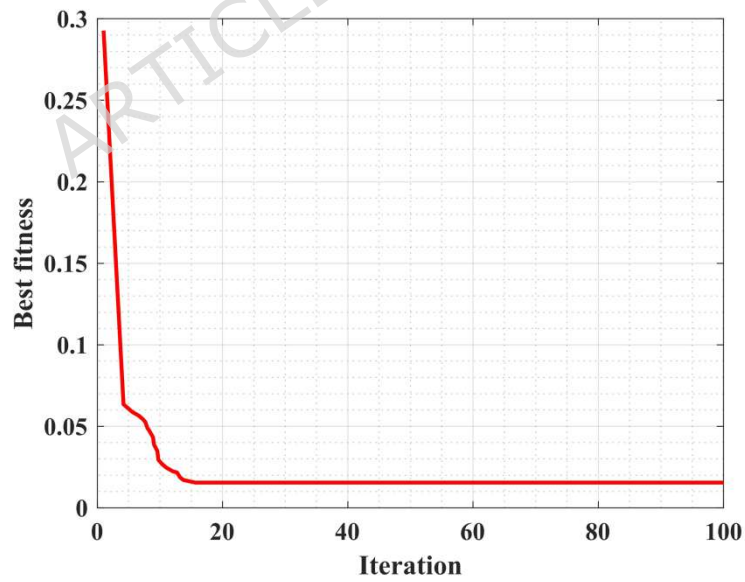


Figure 18: Convergence Analysis of SFOA

The convergence behavior of the SFOA during the 100 iterations of TIFDNFD controller parameter tuning is shown in Figure 18. The starting value of the best fitness is approximately 0.29 meaning that the error at the beginning of the optimization is large. The algorithm quickly finds better controller parameter combinations, as shown by the fitness value rapidly decreasing to almost 0.06 within the first 10 iterations. The fitness value decreases even more by about 20 iterations to about 0.01; following that, the curve progressively flattens and stays near 0 for the remaining iterations. This quick drop and early stabilization show that the SFOA effectively reaches the optimal solution, exhibiting robust exploration capabilities in the early stages and effective exploitation in the later stages. Therefore, the convergence profile demonstrates that SFOA is very effective in reducing error and reaching the best controller tuning in a limited number of iterations.

Table 3: Performance Comparison of Controllers

Controller	Peak Time (s)	Peak Value (rpm)	Peak Overshoot (%)	Settling Time (s)
FOSMC	0.2650	3265	8.81%	0.318
SIIFC-RBFNN	0.0171	3500	16.7%	0.139
TIFDNFD	0.0499	3300	10%	0.208
Proposed	0.0484	3100	3.3%	0.12

Table 3 compares the transient operation of proposed TIFDNFD-SFOA controller and the current controller based on peak time, peak speed, overshoot, and settling time. Although it produces the highest peak value of about 3500 rpm and the largest overshoot of 16.7%, the SIIFC-RBFNN controller has the fastest peak time of 0.0171 s, indicating poor stability. Reaching approximately 3300 rpm with a 10% overshoot, the TIFDNFD controller displays a moderate peak time of 0.0499 s. With a peak speed of 3265 rpm and an overshoot of 8.81%,

the FOSMC controller responds the slowest, reaching its maximum speed at 0.2650 s. Its settling time is the longest, at 0.318 s, indicating slow dynamic behavior. The proposed TIFDNFD-SFOA controller reaches steady-state in 0.12 seconds, which makes it the fastest to settle, and reaches a peak speed of 3100 rpm with a peak time of 0.0484 s, which is almost identical to TIFDNFD but restricts overshoot to just 3.3%. Out of all the compared controllers, the proposed controller provides the best transient performance with the least amount of overshoot and the fastest settling time.

4.10 Discussion

Using the MDIBC converter and the TIFDNFD-SFOA controller, the proposed system's accurately regulate the BLDC motor's voltage and speed. To maintain effective motor operation, the MDIBC raises the input voltage from 250 V to about 480 V, ensuring a smooth and undisturbed DC link voltage to the inverter. The balanced three-phase AC signals can be generated in the inverter, owing to the managed DC link voltage, which enables the easy commutation of the BLDC motor. In comparison to the traditional methods, the suggested TIFDNFD-SFOA controller guarantees quick and precise speed regulation on the motor side, enabling the motor to go from 0 rpm to 3000 rpm in 0.12 seconds with a very low overshoot of 3.3%. The controller minimizes error metrics like ITAE, lowers torque ripple, and improves dynamic behavior during load variations by continuously modifying the control parameters in response to the system response. Therefore, the proposed EV drive system is achieved by the BLDC motor operating at the required speed with maximum stability, efficiency, and smoothness due to the combination of intelligent speed and voltage regulation through TIFDNFD-SFOA controller.

5. Conclusion

This research presented a high-performance BLDC motor drive system that combines an intelligent TIFDNFD controller optimized by the SFOA with a MDIBC. The TIFDNFD-SFOA controller ensures accurate speed and voltage regulation with minimal overshoot, quicker transient response, and lower steady-state error. The MDIBC efficiently reduces ripple current, enhances current sharing, and guarantees a stable boosted DC-link voltage. The SFOA's ability to achieve optimal controller tuning, which leads to smooth torque behavior, reduced ITAE, and improved transient response, is validated by its rapid convergence behavior. According to the simulation, the proposed controller outperformed the existing methods of control, getting a shortest settling time of 0.12 s, a minimum overshoot of 3.3 percent, minimum ITAE and smooth torque. The convergence characteristics provided additional confirmation of SFOA's effectiveness and resilience in determining the best control parameters. The proposed TIFDNFD-SFOA framework improves dynamic performance, stability, and efficiency overall, making it an effective choice for industrial motor-drive applications and advanced EV propulsion. However, the system performance under real-world variations like temperature effects or battery aging has not been tested, and the current work is restricted to hardware validation. In order to improve efficiency and fault tolerance in EV applications, this approach can be further developed in the future by integrating renewable energy sources and adaptive or AI-based predictive controllers on hardware/HIL platforms.

Author Contributions:

Abdullah Alwabli: Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, writing original draft, Writing review & editing

Funding:

The author received no funding for this work

Data Availability Statement:

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Conflicts of Interest:

The authors confirm that no conflict of interest exists about the publication of this article.

References

1. Amoriya, V., Chauhan, R.K., Mittal, S., Sharma, S., Chauhan, K. and Li, H., 2024. Modelling of Multi-Phase Interleaved based Universal Charging Port for Electric Vehicles. *Smart Grids and Sustainable Energy*, 9(1), p.19.
2. Kripalakshmi, T. and Deepa, T., 2023. Design of DC-DC Converters for Electric Vehicle Wireless Charging Energy Storage System. *Power Converters, Drives and Controls for Sustainable Operations*, pp.127-147.
3. Rathore, V., Reddy, S.R.P. and Rajashekara, K., 2022. An isolated multilevel DC–DC converter topology with hybrid resonant switching for EV fast charging application. *IEEE Transactions on Industry Applications*, 58(5), pp.5546-5557.
4. Tandirovic Gursel, A. and Zülfikaroğlu, A., 2024. Modified all-GaN multidevice interleaved boost converter topology for hybrid electrical vehicles and its miniaturization. *Journal of Power Electronics*, 24(12), pp.1879-1888.
5. Sayed, K., El-Zohri, H.H., Ahmed, A. and Khamies, M., 2024. Application of tilt integral derivative for efficient speed control and operation of BLDC motor drive for electric vehicles. *Fractal and Fractional*, 8(1), p.61.
6. Basha, C.H. and Rani, C., 2022. A New single switch DC-DC converter for PEM fuel cell-based electric vehicle system with an improved beta-fuzzy logic MPPT controller. *Soft Computing*, 26(13), pp.6021-6040.

7. Sivamani, D., Sangari, A., Shyam, D., Anto Sheeba, J., Jayashree, K. and Nazar Ali, A., 2023. ANN-Based Maximum Power Point Tracking Control Configured Boost Converter for Electric Vehicle Applications. *Optimization Techniques in Engineering: Advances and Applications*, pp.59-71.
8. Kumar, K., Devi, V.L., Prasad, A., Gali, H.R. and Tiwari, R., 2022. Analysis of a fuel cell-fed BLDC motor drive with a double boost converter for electric vehicle application. In *Smart Grids for Renewable Energy Systems, Electric Vehicles and Energy Storage Systems* (pp. 59-75). CRC Press.
9. Kumar, V., Chenchireddy, K., Sreejyothi, K.R. and Sujatha, G., 2022. Design and Development of Brushless DC Motor Drive for Electrical Vehicle Application. In *AI Enabled IoT for Electrification and Connected Transportation* (pp. 201-217). Singapore: Springer Nature Singapore.
10. Sundaramoorthy, P., Arun, V., Kumar, B.H., Kavali, J. and Balaji, M., 2022. Investigations on novel hybrid reluctance motor for electric vehicle applications. *IEEE Canadian Journal of Electrical and Computer Engineering*, 45(4), pp.454-465.
11. Kim, S.C., Sangam, N., Pagidipala, S. and Salkuti, S.R., 2022. Design and analysis of BLDC motor driver for hybrid electric vehicles. In *Next Generation Smart Grids: Modeling, Control and Optimization* (pp. 297-311). Singapore: Springer Nature Singapore.
12. Mohamed, A.P., Chandrakala, K.V., Balamurugan, S., Subramaniam, U. and Almakhlles, D., 2024. Adaptive maximum power extraction technique in fuel-cell integrated with novel DC-DC converter topology for low-power electric vehicle applications. *Engineering Science and Technology, an International Journal*, 54, p.101723.

13. Shana, L.P., Gudipalli, A., Fahsyar, P.N.A. and Vaithilingam, C.A., 2023, December. Performance Investigation on BLDC Motor and Regenerative Braking with Super-Capacitor for Electric Vehicle Application. In 2023 Innovations in Power and Advanced Computing Technologies (i-PACT) (pp. 1-6). IEEE.
14. Saiteja, P. and Ashok, B., 2024. Development of supervisory self-learning-based energy management controller to control the torque ripples of brushless DC motor in electric vehicles applications. *Arabian Journal for Science and Engineering*, 49(12), pp.16005-16031.
15. Patel, A.N. and Suthar, B.N., 2024. Performance optimisation of axial flux permanent magnet brushless DC motor for electric vehicle application with the genetic algorithm (GA) approach. *International Journal of Ambient Energy*, 45(1), p.2370850.
16. Saha, B., Singh, B. and Sen, A., 2023. SMO based position sensorless BLDC motor drive employing canonical switching cell converter for light electric vehicle. *IEEE Transactions on Industry Applications*, 59(3), pp.2974-2984.
17. Mishra, A.K., Singh, A.K. and Vishwanath, G.M., 2022. A fuel-efficient BLDC motor-driven light electric vehicle with single-stage onboard charging system. *IEEE Transactions on Transportation Electrification*, 9(4), pp.4909-4921.
18. Lins, A.W. and Krishnakumar, R., 2023. Tuning of PID controller for a PV-fed BLDC motor using PSO and TLBO algorithm. *Applied Nanoscience*, 13(4), pp.2911-2934.
19. VK, A.R. and Prasad, V., 2023. Online adaptive gain for passivity-based control for sensorless BLDC motor coupled with DC motor for EV application. *IEEE Transactions on Power Electronics*, 38(11), pp.13625-13634.
20. Kumar, P.H., Lakhimsetty, S. and Somasekhar, V.T., 2023. A low-cost fault-tolerant permanent magnet brush-less direct current motor drive for low-power electric vehicle

- applications. *International Journal of Circuit Theory and Applications*, 51(9), pp.4442-4453.
21. Anbarasi, M.P., Latha, R. and Adharsh Babu, S., 2022. Artificial Neural Network Controlled Multi-Device Interleaved DC-DC Boost Converter Implementation for Electric Vehicle Applications. *IEEE Transactions on Electrical and Electronic Engineering*, 17(10), pp.1420-1431.
 22. Kannan, R. and Sundharajan, V., 2023. A novel MPPT controller based PEMFC system for electric vehicle applications with interleaved SEPIC converter. *International journal of hydrogen energy*, 48(38), pp.14391-14405.
 23. Dhineshkumar, K., Vengadachalam, N., Muthusamy, S. and Khan, B., 2025. Integrated MPPT and bidirectional DC DC converter with reduced switch multilevel inverters for electric vehicles applications. *Scientific Reports*, 15(1), p.25053.
 24. Alzahrani, A., Devarajan, G., Subramani, S., Vairavasundaram, I. and Ogbuka, C.U., 2023. Analysis and validation of multi-device interleaved dc-dc boost converter for electric vehicle applications. *IET Power Electronics*, 16(9), pp.1548-1557.
 25. Shukla, T. and Kalla, U.K., 2022. A BL-CC converter-based BLDC motor drive for marine electric vehicle applications. *International Transactions on Electrical Energy Systems*, 2022(1), p.7026462.
 26. Huang, C.L., Lee, F.C., Liu, C.J., Chen, J.Y., Lin, Y.J. and Yang, S.C., 2022. Torque ripple reduction for bldc permanent magnet motor drive using dc-link voltage and current modulation. *IEEE Access*, 10, pp.51272-51284.
 27. Zhou, X., Sheng, B., Liu, W., Chen, Y., Wang, L., Liu, Y.F. and Sen, P.C., 2021. A high-efficiency high-power-density on-board low-voltage DC–DC converter for electric vehicles application. *IEEE Transactions on Power Electronics*, 36(11), pp.12781-12794.

28. Kumar, A., Naik, M.V. and Kumar, R., 2025. Design and analysis of BLDC motor speed control for electric vehicles powered by solar PV and grid supply. *Discover Electronics*, 2(1), pp.1-33.
29. Venkatesh, R.J., Priya, R., Hemachandu, P. and Reddy, C.V.K., 2024. An optimization approach control of EV solar charging system with step-up DC–DC converter. *Analog Integrated Circuits and Signal Processing*, 119(2), pp.215-232.
30. Tabak, A. and Duman, S., 2024. Maiden application of TID μ 1ND μ 2 controller for effective load frequency control of non-linear two-area power system. *IET Renewable Power Generation*, 18(7), pp.1269-1291.
31. Kumar, M., Tiwari, S., Kumar, V., Sampathi, S.J. and Jarial, R.K., 2024, May. Implementation of Field-Oriented Control (FOC) Algorithm for Brushless DC (BLDC) Motor Speed Regulation. In 2023 Second IEEE International Conference on Measurement, Instrumentation, Control and Automation (ICMICA) (pp. 1-6). IEEE.
32. Jia, H., Zhou, X., Zhang, J. and Mirjalili, S., 2025. Superb Fairy-wren Optimization Algorithm: a novel metaheuristic algorithm for solving feature selection problems. *Cluster Computing*, 28(4), p.246.
33. Alnaib, I.I., Alsammak, A.N. and Mohammed, K.K., 2025. Brushless DC motor drive with optimal fractional-order sliding-mode control based on a genetic algorithm. *Electrical Engineering & Electromechanics*, (2), pp.19-23.